🟢 DAY 1 — SQL Fundamentals

Questions 1–15

Revise: SELECT, WHERE, DISTINCT, ORDER BY, LIMIT, operators,

NULL, aliases.

Employee Data Queries

1. Display all employees.
2. Display employee names and salaries.
3. Find employees earning more than ₹50,000.
4. Find employees belonging to the IT department.
5. Find employees whose salary is between ₹40,000 and ₹80,000.
6. Find employees whose names start with A.
7. Find employees whose names end with n.
8. Find employees whose name contains "an".
9. Display unique department names.
10. Sort employees by salary ascending.
11. Sort employees by salary descending.
12. Find the top 5 highest-paid employees.
13. Find employees who don't have a manager.
14. Find employees whose salary is not NULL.
15. Find employees hired after 2022.

🎯 Goal: Be extremely comfortable writing basic queries without

thinking about syntax.

🟢 DAY 2 — Aggregation&GROUP BY

Questions 16–30

Revise: COUNT, SUM, AVG, MIN, MAX, GROUP BY, HAVING.

Employee Data Analysis Queries

I. Company-Wide Metrics

1. Find the total number of employees.
2. Find the average salary.
3. Find the maximum salary.
4. Find the minimum salary.
5. Find the total salary paid by the company.

II. Departmental Metrics

1. Count employees in each department.
2. Find average salary by department.
3. Find maximum salary in each department.
4. Find minimum salary in each department.
5. Find total salary by department.

III. Departmental Filtering and Ranking

1. Find departments having more than 10 employees.
2. Find departments with average salary above ₹70,000.
3. Find the department with the highest average salary.
4. Find the department with the lowest average salary.
5. Find the second-highest average departmental salary.

🔥 Interview trap: Know exactly when to use WHERE vs HAVING.

🟡 DAY 3 — JOINs

Questions 31–45

Revise: INNER JOIN, LEFT JOIN, RIGHT JOIN, self join, multiple joins.

Assume:

employees(employee_id, name, department_id, manager_id, salary)

departments(department_id, department_name)

Display employee names with department names.

Display all employees, including those without departments.

Employee and Department Queries

1. Find employees working in the IT department.
2. Find departments having no employees.
3. Display employee name and manager name.
4. Find employees earning more than their managers.
5. Find employees earning less than their managers.
6. Find the number of employees in each department.
7. Find average salary by department.
8. Find employees who don't have a matching department.
9. Join three tables: employees, departments, and locations.
10. Find employees working in Bangalore.
11. Find departments with average salary above ₹60,000.
12. Find the highest-paid employee in each department using a JOIN.

13. Find employees who belong to the same department as "John".

💀 Important: If you master JOINs + aggregation, you're already dangerous in SQL interviews.

🟡 DAY 4 — Subqueries, CTEs & CASE

Questions 46–60

Revise: subqueries, correlated subqueries, CTE, CASE, EXISTS, IN.

SQL Query Examples

1. Find employees earning more than the average salary.
2. Find the employee with the highest salary.
3. Find the second-highest salary.
4. Find employees earning the second-highest salary.
5. Find employees earning more than the average salary of their department.
6. Find departments whose average salary is above company average.
7. Find employees working in the same department as "Alice".
8. Find employees who earn more than every employee in HR.
9. Find employees who earn more than at least one employee in HR.
10. Find departments containing the highest-paid employee.
11. Categorize employees as Low/Medium/High salary using CASE.
12. Create a CTE containing department-wise average salary.
13. Use a CTE to find departments above company average.

14. Find employees who have never made a sale using NOT EXISTS.

15. Find customers who have placed at least one order using EXISTS.

🔥 Focus: Don't memorize subquery patterns. Understand why the inner query exists.

🔴 DAY 5 — Window Functions

Questions 61–75

This is VERY important for Data Analyst/Data Scientist interviews.

Revise:

ROW_NUMBER()

RANK()

DENSE_RANK()

LAG()

LEAD()

SUM() OVER()

AVG() OVER()

PARTITION BY

Employee and Sales Analysis

Employee Analysis

1. Rank employees by salary.

2. Rank employees within each department.

3. Find the highest-paid employee in each department.

4. Find the second-highest-paid employee in each department.

5. Find the top 3 employees in each department.

6. Assign row numbers to employees.

7. Find duplicate records using ROW_NUMBER().

Order and Sales Analysis

1. Find the latest order for each customer.

2. Find the first order for each customer.

3. Calculate running total of sales.

4. Calculate running average of sales.

5. Find each employee's salary difference from the department

average.

6. Find each employee's salary difference from the previous employee.

7. Compare current month's sales with previous month's sales using

LAG().

8. Calculate month-over-month sales growth.

🚨 Golden rule:

ROW_NUMBER ≠ RANK ≠ DENSE_RANK

You should be able to explain the difference without hesitation.

🔴 DAY 6 — Real-World SQL Problems

Questions 76–90

Data Analyst / Data Scientist Interview Scenarios

1. **Customer Record Management:**

   Find duplicate customer records.

Remove duplicate records while keeping the latest record.

2. **Customer Order Analysis:**

Find customers who haven't ordered anything.

Find customers with more than 3 orders.

Find each customer's first purchase date.

Find each customer's most recent purchase.

Find customers who purchased in both January and February.

Find customers whose spending increased compared with the previous month.

3. **Product Performance Analysis:**

Find the highest-selling product.

Find the top 3 products by revenue.

4. **Revenue Analysis:**

Find monthly revenue.

Find yearly revenue.

Find the month with the highest revenue.

Find customers whose total spending exceeds ₹1 lakh.

Find the average order value.

🔥 These are the questions where your SQL starts becoming job-ready.

🏆 DAY 7 — Mock Interview Day

Questions 91–100

Today: NO notes.

Set a timer for 90 minutes.

Imagine an interviewer is sitting in front of you.

SQL Query Challenges

1. Find the second-highest salary without using LIMIT.
2. Find the Nth-highest salary.
3. Find the top 3 salaries in every department.
4. Find employees whose salary is higher than their department average.
5. Find duplicate records and keep only the latest record.
6. Find the longest consecutive purchasing streak of a customer.
7. Find customers who purchased every month in a given year.
8. Calculate month-over-month revenue growth.
9. Find the highest-revenue product in each category.

Case Study: Customer Report

Given Customers, Orders, and Products tables, produce a report containing:

- Customer name
- Total orders
- Total spending
- Average order value
- First purchase date

Latest purchase date

Customer spending rank